3Chapters 1, 2 and 3

Review Questions chapters 1, 2 and 3

Chapter 1 - Test Questions

How does software differ from the artifacts produced by other engineering disciplines?
Software and artifacts from other engineering fields like a car engine or bridge from mechanical engineering, a circuit board from electrical engineering have big differences. Main distinction is that software is intangible and changeable. It won't wear out because it doesn't degrade over time like a car engine. When it stops it's because of a design flaw or bug not from physical use. And Software can be very complex and doesn't always follow strict, universal standards like those used for building materials or safety in physical engineering.

How do software characteristics differ from hardware characteristics?

For example, smartphones are examples of hardware. It is a physical phone itself with its screen, battery and microchip. On the other hand, software is an operation system, apps and games that run on.
So, the software is intangible, does not wear out, is easily modifiable and costly to develop but chops to reproduce. Hardware is tangible, subject to physical wear, harder to modify once built and incurs material and Manufactuing costs for each unit.

Explain what is wrong with the notion that computer software does not need to evolve over time.
Software doesn't exist in a vacuum. It operates within an ever-changing environment of user needs, business requirements, and technology. If the software program is not updated to keep pace with these changes, it will quickly become obsolete and lose its value.

List three areas in which process models may differ from one another.
Process model is a framework that defines the sequence of activities and tasks in software development, guiding how a software product is planned, built, tested and maintained.
Different models exist depending on project needs.
Waterfall Model: sequential with each phase completed before the next one begins. But it's not flexible and doesn't handle changes well. Customer feedback is only given at the end.

Prototyping Model: creates a working model of the system early in the development cycle. It reduces the risk of requirement changes and product rejection because the customer is involved early and often.
Spiral Model: risk-driven and combines elements of both waterfall and prototyping. Excellent for managing development risks and works for large complex and extensible products. But a failure in the risk analysis can doom the entire project.
Unified Process Model: Iterative and incremental, focusing on architecture and use cases. It emphasizes high-quality documentation and accommodates changes in requirements. There is continuous customer involvement, and it works well for maintenance projects.

Waterfall models follow a strict linear sequence, however Agile and spiral are iterative.
Waterfall models get customer feedback at the end However Prototyping spiral models, the customer is a constant partner, providing continuous feedback throughout the entire development process.
Waterfalls are very rigid. On the other hand, agile and spiral are designed to be flexible. They can easily adapt to new requirements that may arise during the project.

Describe how Polya's problem solving principles describe the essence of engineering practice?
->Understand the problem (communication and analysis)
-> Plan a solution (modeling and software design)
-> carryout the plan (code implementation)
-> Examine the result for accuracy (testing and QA)
Polya's problem solving principle describes that the first step is to fully understand the problem and engineers begin by gathering and analyzing requirements. Next step is planning a solution by modeling and designing phases in engineering. This could be a system architecture, a circuit diagram, or a set of software algorithms.
The third step is implementation. Engineers execute the plan they designed. The Final step is testing and reflection. Together, they capture the systematic and iterative nature of engineering.

Chapter 2 - Test Questions

How does software team choose the task set for a particular project?

Choose a task set that best accommodates the needs of the project and characteristics of the team, size of the project and complexity.

How can process patterns assist a development team build software products efficiently?
Process patterns must be coupled with solid software engineering practice In addition, the process itself can be assessed to ensure that it meets a set of basic process criteria that have been shown to be essential for successful software engineering.

Why it important for software processes to be agile?
It's crucial for software processes to be agile because of the fast-faced and unpredictable nature of the software industry. Adapting rapid incremental delivery of software will be essential for creating successful software products with frequent customer feedback.

Describe the phases of the prototyping model for software development?
The phases of prototyping model is iterative loop until product requirements are clear and can be built. First, team gathers and analysis requirements next, creates simple prototype design with visible features. Third, build a working model based on prototype design able to simulate the key features. Forth, the customer provides feedback on the design and features and usability. Fifth, the team refines the prototype based on the feedback. Sixth, once the prototype is approved and requirements are solidified, the team develops the final robust product and test and deploys.

Why are evolutionary models considered by many to be the best approach to software development in a modern context?
Evolutionary models are an unbrella term for development approaches that build software in small, iterative cycles, allowing the product to evolve over time.
Evolutionary models
Spiral model – risk-driven approach that combines prototyping and waterfall.
Team builds a new prototyping or version in each loop analyzing and mitigating risks before moving on. Ideal for large, complex high-risk projects.

Prototyping model - build prototype software early in the development cycle. Gets some feed back form customers and refines it for final product.

Agile model – adaptability and customer collaboration. It delivers working software frequently in small increments. (scrum, unified process)

Chapter 3 - Test Questions

List the key issues stressed by an agile philosophy of software engineering.
Difficulty in predicting customer needs and requirements.
How much design will be needed for the project
Not predictable activities of analysis, design, construction and testing

What are the tradeoffs proposed by the "Manifesto for Agile Software Development"?
Communication and collaboration are more important than processes and tools
Getting a valuable working software is more crucial than documentation
Instead of rigid contracts, frequent close collaboration with the customer throughout the project is important.
Requirements of the project is changeable but the ability to adapt the change is valued .
Describe the role of customers and end-users on an agile process team?
Customers offer feedback through the overall process. And end-users provide detailed practical feedback for product's usability and functionality.

Describe the three key assumptions regarding software projects that every agile software process must address.
First, it is difficult to predict the changeability of software requirements and customer's needs. Agile process welcomes iterative continuous feedback from customers.
Second, it is hard to predict how much design is necessary before construction. Agile process promotes an iterative and incremental approach to design.
Third, analysis, design, construction and testing are not predictable. So, in agile process all the activities are compressed in timeframe iteratively.

What are the three questions that should be answered by each team member at the daily Scrum meeting?
What did you do since the last team meeting?
What obstacles are you encountering?
What do you plan to accomplish by the next team meeting?